Sets, Set Operations, Functions

Dr. Nafis Sadeq
Assistant Professor
Department of Computer Science and Engineering
East West University



Set Basics

Definition of a Set

Definition

A **set** is an unordered collection of objects, called *elements* or *members* of the set. A set is said to *contain* its elements.

We write $a \in A$ to denote that a is an element of the set A. The notation $a \notin A$ denotes that a is not an element of A.

- Sets are fundamental objects in mathematics used to group elements together.
- The order of elements in a set does not matter.

Example Sets

- The set of all vowels in the English alphabet:

$$V = \{a, e, i, o, u\}$$

- The set of positive integers less than 100:

$$\{1, 2, 3, \ldots, 99\}$$

- The set of odd positive integers less than 10:

$$O = \{1, 3, 5, 7, 9\}$$

Representing a Set

There are two standard ways to represent a set:

1. **Roster Form** List all elements explicitly:

$$O = \{1, 3, 5, 7, 9\}$$

2. **Set-Builder Form** Describe the property that defines all members:

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

or equivalently,

$$O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}$$

Commonly Used Set Notations

- $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$
  the set of **natural numbers**
- $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$
  the set of **integers**
- $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$
  the set of **positive integers**
- $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$
  the set of **rational numbers**

- $\mathbb{R}$: set of **real numbers**
- $\mathbb{R}^+$: set of **positive real numbers**
- $\mathbb{C}$: set of **complex numbers**

Equality of Sets

Definition

Two sets A and B are said to be **equal** if and only if they contain exactly the same elements:

$$A = B \iff (\forall x)(x \in A \Leftrightarrow x \in B)$$

- The order of elements does not matter.

$$\{1, 2, 3\} = \{3, 2, 1\}$$

- Repetition of elements does not affect equality.

$$\{1, 2, 2, 3\} = \{1, 2, 3\}$$

Universal and Empty Set

Definition: Universal Set

The **universal set**, denoted by U, is the set that contains all objects under consideration. All other sets are subsets of U.

Example: $A = \{1, 2, 3, 4, 5\}, B = \{2, 4, 6, 8, 10\}, C = \{3, 5, 7, 9\}$

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Definition: Empty Set

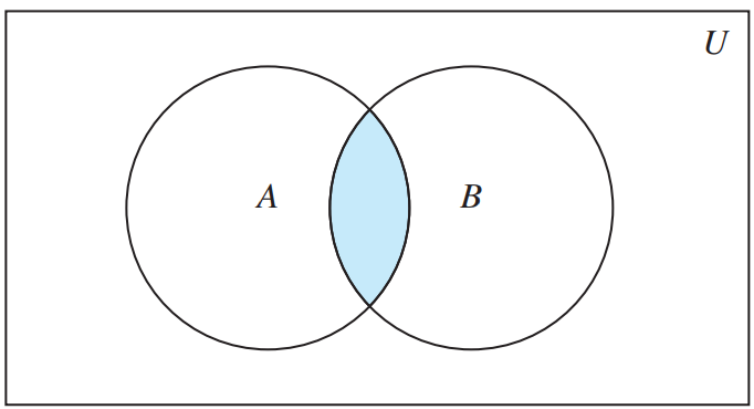
The **empty set** (or **null set**), denoted by $\emptyset$ or $\{\}$, is the set that contains no elements.

$$\emptyset = \{\}$$

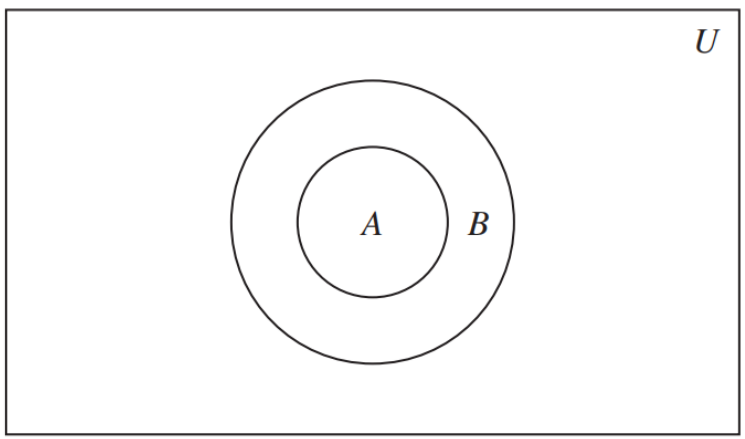
- Every set is a subset of the universal set.
- The empty set is a subset of every set.

Venn Diagram Representation

- **Venn diagrams** are used to visualize relationships among sets.
- The universal set U is represented by a rectangle.
- Each subset is represented by a closed curve (often a circle) within U.



$A \cap B$ is shaded.



Subsets

Definition

The set A is a **subset** of B if and only if every element of A is also an element of B. We write:

$$A \subseteq B$$

$$A \subseteq B \iff (\forall x)(x \in A \to x \in B)$$

- To show $A \not\subseteq B$, find one element $x \in A$ such that $x \notin B$.
- Every nonempty set S has at least two subsets:

$$\emptyset \subseteq S \quad \text{and} \quad S \subseteq S$$

Cardinality of a Set

Definition

Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, then S is a **finite set**, and n is the **cardinality** of S.

$$|S| = n$$

- **Example 1:** $A = \{1, 3, 5, 7, 9\} \Rightarrow |A| = 5$
- **Example 2:** $S = \{\text{all English letters, not case sensitive}\} \Rightarrow |S| = 26$
- **Example 3:** $\emptyset$ has no elements $\Rightarrow |\emptyset| = 0$

Power Sets

Definition

Given a set S, the **power set** of S, denoted by $\mathcal{P}(S)$, is the set of all subsets of S.

- If S has n elements, then $|\mathcal{P}(S)| = 2^n$.
- The power set always includes both $\emptyset$ and S itself.

Example:

$$S = \{0, 1, 2\}$$

$$\mathcal{P}(S) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$$

Definition: Cartesian Product

Definition

Let A and B be sets. The **Cartesian product** of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

- The elements of $A \times B$ are ordered pairs.
- The order matters: $(a, b) \neq (b, a)$ unless $a = b$.
- If either A or B is empty, then $A \times B = \emptyset$.

Example: Cartesian Product of Two Sets

Given:

$$A = \{1, 2\}, \quad B = \{a, b, c\}$$

Then:

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

- Each element of A is paired with every element of B.
- Total number of ordered pairs: $|A \times B| = |A| \times |B| = 2 \times 3 = 6$.
- Note that $A \times B \neq B \times A$ (order matters).

Visualizing Cartesian Product

- Each element of A (horizontal axis) pairs with every element of B (vertical axis).
- The result forms a rectangular grid of ordered pairs.

$$A = \{1, 2\}, \quad B = \{a, b, c\}$$

$A \backslash B$	a	b	c
1	(1,a)	(1,b)	(1,c)
2	(2,a)	(2,b)	(2,c)

Example: Cartesian Product of Three Sets

Given:

$$A = \{0, 1\}, \quad B = \{1, 2\}, \quad C = \{0, 1, 2\}$$

Then:

$$A \times B \times C = \{(a, b, c) \mid a \in A, b \in B, c \in C\}$$

Explicitly:

$$\begin{aligned} A \times B \times C = \{&(0,1,0), (0,1,1), (0,1,2), \\ &(0,2,0), (0,2,1), (0,2,2), \\ &(1,1,0), (1,1,1), (1,1,2), \\ &(1,2,0), (1,2,1), (1,2,2)\}. \end{aligned}$$

- Total ordered triples: $|A \times B \times C| = 2 \times 2 \times 3 = 12$.

Set Operations

Set Union

Definition

Let A and B be sets. The **union** of A and B, denoted by $A \cup B$, is the set that contains all elements that are in A, or in B, or in both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

- The union combines the elements of both sets without repetition.
- The operation is commutative: $A \cup B = B \cup A$

Example

$$A = \{1, 3, 5\}, \quad B = \{1, 2, 3\}$$

$$A \cup B = \{1, 2, 3, 5\}$$

Set Intersection

Definition

Let A and B be sets. The **intersection** of A and B, denoted by $A \cap B$, is the set that contains those elements common to both A and B.

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

- The intersection contains only elements that appear in both sets.
- The operation is commutative: $A \cap B = B \cap A$

Example

$$A = \{1, 3, 5\}, \quad B = \{1, 2, 3\}$$

$$A \cap B = \{1, 3\}$$

Disjoint Sets

Definition

Two sets A and B are said to be **disjoint** if their intersection is the empty set:

$$A \cap B = \emptyset$$

Example

$$A = \{1, 3, 5, 7, 9\}, \quad B = \{2, 4, 6, 8, 10\}$$

$$A \cap B = \emptyset$$

Hence, A and B are disjoint.

Set Difference

Definition

Let A and B be sets. The **difference** of A and B, denoted by $A - B$, is the set containing those elements that are in A but not in B.

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$

- $A - B$ is also called the *complement of B with respect to A*.
- Sometimes denoted by $A \setminus B$.

Example

$$A = \{1, 3, 5\}, \quad B = \{1, 2, 3\}$$

$$A - B = \{5\}, \quad \text{while } B - A = \{2\}.$$

Set Complement

Definition

Let U be the universal set. The **complement** of A, denoted by $\overline{A}$, is the set of all elements in U that are not in A.

$$\overline{A} = \{x \in U \mid x \notin A\}$$

- $\overline{A} = U - A$
- $A \cup \overline{A} = U$ and $A \cap \overline{A} = \emptyset$

Example

If $U = \{1, 2, 3, 4, 5\}$ and $A = \{1, 3, 5\}$, then

$$\overline{A} = \{2, 4\}.$$

Set Identities

Set identities are algebraic relationships among set operations. They are analogous to algebraic laws in arithmetic and logic.

Identity	**Name**
$A \cap U = A,\ A \cup \emptyset = A$	Identity laws
$A \cup U = U,\ A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A,\ A \cap A = A$	Idempotent laws
$\overline{\overline{A}} = A$	Double Complement law
$A \cup (A \cap B) = A,\ A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U,\ A \cap \overline{A} = \emptyset$	Complement laws

Important Set Laws

- **Commutative Laws:**

$$A \cup B = B \cup A, \quad A \cap B = B \cap A$$

- **Associative Laws:**

$$A \cup (B \cup C) = (A \cup B) \cup C$$

- **Distributive Laws:**

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

- **De Morgan's Laws:**

$$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad \overline{A \cap B} = \overline{A} \cup \overline{B}$$

Example: Union and Intersection of Three Sets

Given:

$$A = \{0, 2, 4, 6, 8\}, \quad B = \{0, 1, 2, 3, 4\}, \quad C = \{0, 3, 6, 9\}$$

Find: $A \cup B \cup C$ and $A \cap B \cap C$

Solution:

$$A \cup B \cup C = \{0, 1, 2, 3, 4, 6, 8, 9\}$$

$$A \cap B \cap C = \{0\}$$

- The union includes all elements from any of the sets.
- The intersection includes only elements common to all three sets.

Functions

Definition of a Function

Function

Let A and B be nonempty sets. A **function** f from A to B is an assignment of exactly one element of B to each element of A.

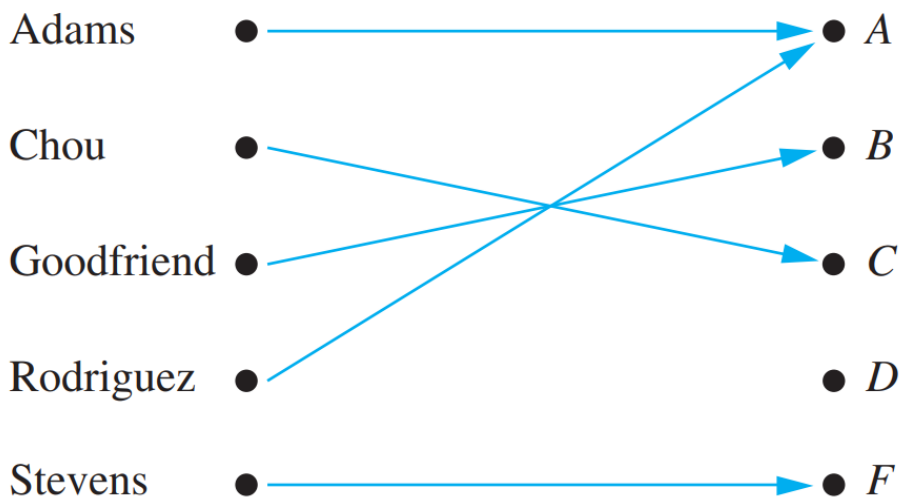
$$f(a) = b \text{ if } b \text{ is the unique element of } B \text{ assigned by } f \text{ to } a.$$

If f is a function from A to B, we write:

$$f : A \to B$$

- Each input $a \in A$ has a unique output $b \in B$.
- A function can be visualized as a mapping between two sets.

Example: Function Mapping Visualization



$A = \{\text{Adams, Chou, Goodfriend, Rodriguez, Stevens}\}$

$$B = \{A, B, C, D, F\}$$

Each student is assigned exactly one grade.

Domain, Codomain, and Range

Definitions

If $f : A \to B$ is a function:

- A is the **domain** of f.
- B is the **codomain** of f.
- The **range (or image)** of f is the set of all images of elements of A.

If $f(a) = b$, we say:

- b is the **image** of a,
- a is a **preimage** of b.

Example: Grades as a Function

Problem: What are the domain, codomain, and range of the function that assigns grades to students?

Solution: Let G be the function assigning a grade to each student in a class.

$$\text{Domain of } G = \{\text{Adams, Chou, Goodfriend, Rodriguez, Stevens}\}$$

$$\text{Codomain of } G = \{A, B, C, D, F\}$$

$$\text{Range of } G = \{A, B, C, F\}$$

Since each grade except D is assigned to at least one student.

Operations on Functions

Sum and Product of Functions

Let f_1 and f_2 be functions from A to $\mathbb{R}$. Then new functions $f_1 + f_2$ and $f_1 f_2$ from A to $\mathbb{R}$ are defined by:

$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$

$$(f_1 f_2)(x) = f_1(x) f_2(x)$$

- Function operations are performed by combining outputs for the same input.
- These definitions extend naturally to other algebraic operations.

Example: Sum and Product of Functions

Let

$$f_1(x) = x^2, \quad f_2(x) = x - x^2$$

Find $f_1 + f_2$ and $f_1 f_2$.

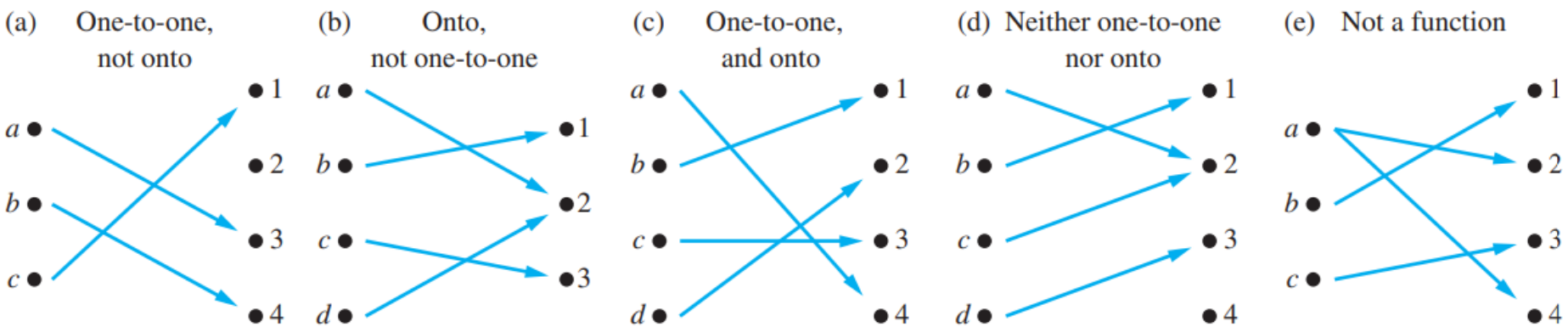
Solution:

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

$$(f_1 f_2)(x) = f_1(x) f_2(x) = x^2(x - x^2) = x^3 - x^4$$

- The sum results in a linear function.
- The product results in a polynomial function of degree 4.

Types of Functions



One-One (Injective) Function

Definition

A function f is said to be **one-to-one** or an **injection** if and only if

$$f(a) = f(b) \implies a = b$$

for all a, b in the domain of f. Equivalently, f is one-to-one if and only if $f(a) \neq f(b)$ whenever $a \neq b$.

- Alternate term: **Injective function**.
- Each element in the codomain is the image of at most one element in the domain.

Examples: One-One Function

Example 1: Determine whether $f(x) = x^2$ (from integers to integers) is one-to-one.

Solution:

$$f(1) = 1, \quad f(-1) = 1 \quad \text{but } 1 \neq -1$$

Hence, $f(x) = x^2$ is **not one-to-one** over $\mathbb{Z}$.

However, if we restrict the domain to $\mathbb{Z}^+$, $f(x) = x^2$ **becomes one-to-one**.

Example 2: $f(x) = x + 1$ over real numbers is one-to-one because

$$x + 1 \neq y + 1 \text{ whenever } x \neq y.$$

So $f(x) = x + 1$ **is one-to-one**

Onto (Surjective) Function

Definition

A function $f : A \to B$ is said to be **onto**, or a **surjection**, if and only if for every element $b \in B$, there exists an $a \in A$ such that

$$f(a) = b$$

- Alternate term: **Surjective function**.
- Every element of the codomain B is an image of some element in the domain A.

Examples: Onto Function

Example 1: Is $f(x) = x^2$ (over real numbers) onto?

Solution: No, because there is no integer x such that $x^2 = -1$. Hence, $f(x) = x^2$ is **not onto**.

Example 2: Is $f(x) = x + 1$ (from integers to integers) onto?
Solution: Yes, because for any integer y, there exists $x = y - 1$ such that $f(x) = y$. Hence, $f(x) = x + 1$ is **onto**.

Bijective Function

Definition

A function $f : A \to B$ is called **bijective** if it is both:

- One-to-one (injective)
- Onto (surjective)

- Every element in B is mapped from exactly one element in A.
- A bijective function establishes a **one-to-one correspondence** between sets A and B.
- Such functions have an **inverse function** $f^{-1} : B \to A$.

Examples: Bijective Function

Example: Let $f(x) = x + 1$ from $\mathbb{Z}$ to $\mathbb{Z}$.

Solution:

- It is one-to-one since $x + 1 = y + 1 \implies x = y$.
- It is onto since for every $y \in \mathbb{Z}$, $x = y - 1$ exists.

Therefore, $f(x) = x + 1$ is a **bijective function**.

Inverse Function — Definition

Definition

Let f be a one-to-one correspondence from a set A to a set B. The **inverse function** of f, denoted f^{-1}, assigns to each element $b \in B$ the unique element $a \in A$ such that

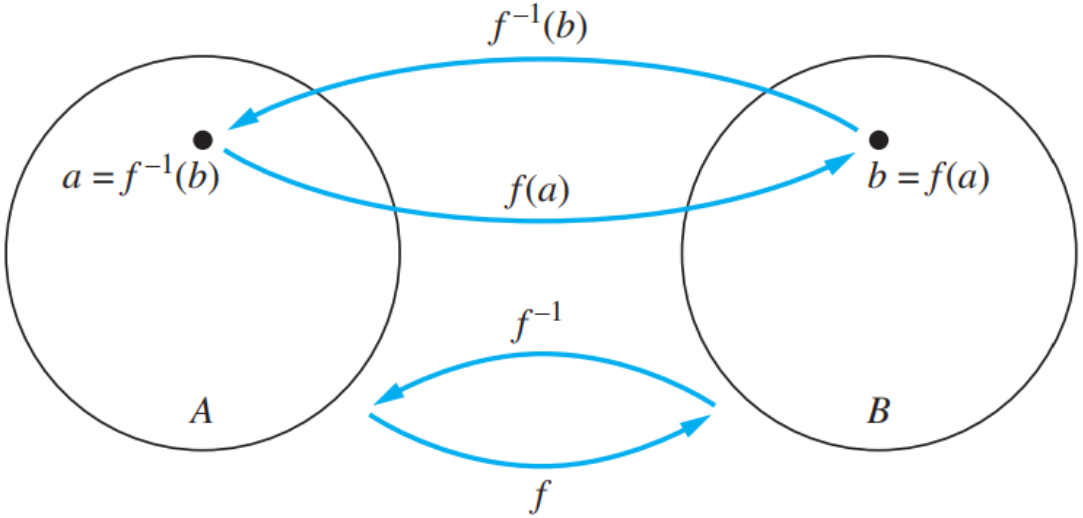
$$f(a) = b.$$

Thus,

$$f^{-1}(b) = a \quad \text{when } f(a) = b.$$

- f^{-1} reverses the mapping direction of f.
- Only **bijective functions** have inverses.

Inverse Function — Visualization



$$f : A \to B, \quad f^{-1} : B \to A$$

$$f(a) = b \iff f^{-1}(b) = a$$

Examples of Inverse Functions

Example 1: Let $f : \mathbb{Z} \to \mathbb{Z}$ be defined by $f(x) = x + 1$. Is f invertible, and what is its inverse?

Solution: Since f is one-to-one and onto, it has an inverse. Let $y = f(x) = x + 1 \Rightarrow x = y - 1$. Hence,

$$f^{-1}(y) = y - 1.$$

Example 2: Let $f(x) = x^2$ from $\mathbb{R} \to \mathbb{R}$. This function is **not invertible** because $f(-2) = f(2) = 4$. It fails the one-to-one property.

Composite Function — Definition

Definition

Let $g : A \to B$ and $f : B \to C$. The **composition** of f and g, denoted $f \circ g$, is defined by:

$$(f \circ g)(a) = f(g(a))$$

for all $a \in A$.

- The output of g becomes the input of f.
- The order of composition matters: $f \circ g \neq g \circ f$.

Example: Composite Functions

Let

$$f(x) = 2x + 3, \quad g(x) = 3x + 2.$$

Find: $(f \circ g)(x)$ and $(g \circ f)(x)$

Solution:

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11$$

Observation: $f \circ g \neq g \circ f$

Example: Computing Composite Values

Given: $f(x) = 3x + 2, \quad g(x) = x^2 + 1$

Compute:

$$(f \circ g)(2), \quad (g \circ f)(2)$$

Solution:

$$(f \circ g)(2) = f(g(2)) = f(2^2 + 1) = f(5) = 3(5) + 2 = 17$$

$$(g \circ f)(2) = g(f(2)) = g(3(2) + 2) = g(8) = 8^2 + 1 = 65$$

Result: $(f \circ g)(2) = 17, \ (g \circ f)(2) = 65$

Floor and Ceiling Functions — Definition

Definition

For any real number x:

$$\lfloor x \rfloor = \text{the greatest integer less than or equal to } x$$

$$\lceil x \rceil = \text{the smallest integer greater than or equal to } x$$

- $\lfloor 3.7 \rfloor = 3, \quad \lceil 3.7 \rceil = 4$
- $\lfloor -2.3 \rfloor = -3, \quad \lceil -2.3 \rceil = -2$